ACTIVITY 1

What is required?

A small glass bottle, a rubber cork, an empty refill, water, colour, a candle, a fork, a paper.

What to do?

- Take a small glass bottle. Fill it with coloured water.
- Make hole at the centre of the rubber cork.
- Pass empty refill from the hole of the rubber cork.
- Make the bottle air tight and observe the water raised in the refill.
- Make a scale on paper, place it behind the refill and note down the position of the surface of water.
- Hold bottle with fork and supply heat to it with candle. Then observe.



What is the change in the surface of water?

- Stop the supply of heat. When water is cooled, observe the surface of water in the refill, **what change takes place? Why?**

When, a liquid is heated, it expands and when it is cooled down, it contracts.

Properties of Mercury:-

- Its expansion is uniform. (For equal amounts of heat it expands by equal lengths.)
- It is opaque and shining.
- It does not stick to the sides of the glass tube.
- It is a good conductor of heat.
- It has a high boiling point (357°C) and a low freezing point (–39°C). Hence a wide range of temperatures can be measured using a mercury thermometer

Properties of Alcohol

- The freezing point of alcohol is less than –100°C. So it can be used to measure very low temperatures.
- Its expansion per degree Celsius rise in temperature is very large.
- It can be coloured brightly and hence is easily visible.

1.4 Types of Thermometers

There are different types of thermometers for measuring the temperatures of different things like air, our bodies, food and many other things. Among these, the commonly used thermometers are clinical thermometers and laboratory thermometers.

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1.4.1 Clinical Thermometer

These thermometers are used to measure the temperature of a human body, at home, clinics and hospitals. All clinical thermometers have a kink that prevents the mercury from

VII Std Science Term-2 EM Unit 1.indd 3 20-07-2019 1.46.39 PM